Did Liquidity Govern Stablecoin Peg Pass-Through? Evidence from the March 2023 USDC Depeg

Samson Lam

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Stablecoin depegs and arbitrage

- A stablecoin is a cryptocurrency that trades at a fixed target against a fiat currency, one US dollar (USD) for Tether (USDT) and USD Coin (USDC). The stablecoin price of Bitcoin (BTC) is a dollar price only while the peg holds.
- Perez Riaza and Gnabo (2025) find that a Tether depeg raises the probability of a jump in BTC/USD almost fivefold within 5 minutes. Bartůněk (2024) finds that depegging raises the realized volatility of Bitcoin.
- Makarov and Schoar (2020) document large arbitrage deviations across crypto exchanges. Reynolds et al. (2021) find deviations from triangular arbitrage parity in Bitcoin markets but not among fiat currencies.
- Lyons and Viswanath-Natraj (2023) show that Tether's peg deviations halved once arbitrage with the issuer became easier.

Research question

Question

Did liquidity govern how stablecoin peg stress passed into the stablecoin price of Bitcoin during the March 2023 USDC depeg?

Finding

1. During the crisis, the peg deviation carried most of the basis variance, and the triangular arbitrage deviation had a half-life below 1 minute.
2. Liquidity did not measurably govern the pass-through.
3. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.
4. The stress that reached the stablecoin price of Bitcoin was the peg itself, that arbitrage relationships were maintined.

Sample and notation

$$F_t = \mathbf{1}\{v_t^{\mathrm{BTC/USD}} > 0,\ v_t^{\mathrm{BTC}/Q} > 0,\ v_t^{Q/\mathrm{USD}} > 0\}, \qquad FF_t = F_t F_{t-1}$$

- We use 1-minute data from Binance.US for five pairs, BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD, from 1 to 21 March 2023.
- Write $P_t^{X/Y}$ for the close price of one unit of X in units of Y at time t, and $Q \in \{\mathrm{USDT}, \mathrm{USDC}\}$ for a stablecoin.
- Deviations are in basis points (bps), so log prices are scaled by $m = 10^4$.
- F_t therefore indicates that all three pairs of the triangle trade at time t, and FF_t that both t and $t-1$ have $F = 1$, with v the pair volume.

Depeg event identification

- Following Perez Riaza and Gnabo (2025), π_k is the 5-minute Q/USD close.
- A bar is a breach when $|\pi_k - 1| \geq \tau$, with $\tau = 0.5\%$ in the baseline, as in Perez Riaza and Gnabo (2025) and Polizu et al. (2023).
- An event starts at the last bar before the breach when the price equals 1 dollar exactly, or at the first breach bar when the previous bar is at the peg.
- Estimates for one stablecoin use its own window, and estimates that use all pairs use the union of the two windows, which equals the USDT window.

Stablecoin	Threshold and start	Start	End	Hours
USDT	$\pm 0.5\%$, last-peg start (baseline)	10 Mar 21:10	15 Mar 13:10	112.0
USDC	$\pm 0.5\%$, last-peg start (baseline)	11 Mar 01:20	14 Mar 02:30	73.2

Depeg windows from the 5-minute Q/USD close. Times are UTC in March 2023.

Liquidity measures

$$\text{Kyle}_t = m \left(\frac{\frac{1}{5} \sum_{i=0}^{4} r_{t-i}^2}{\sum_{i=0}^{4} V_{t-i}} \right)^{1/3}$$

- We measure illiquidity with the Kyle estimator of Kyle and Obizhaeva (2016), where r_t is the 1-minute log return and V_t the quote volume of a pair.
- Kyle_t rises with volatility per unit of volume.

Liquidity by pair and regime

	Kyle			Quote volume per day		
Pair	pre	crisis	post	pre	crisis	post
BTC/USD	0.79	1.02	0.88	141	254	333
BTC/USDT	1.14	1.40	1.73	55.4	113	58.1
BTC/USDC	3.08	6.37	5.33	4.15	5.38	3.71
USDT/USD	0.91	1.48	1.44	3.21	8.99	3.89
USDC/USD	1.23	7.23	3.79	1.13	9.76	0.94

Liquidity by pair and regime over the union window. Quote volume is in millions of the quote currency.

- The crisis affected the USDC pairs most.
- The Kyle illiquidity of USDC/USD rose $1.23 \to 7.23$ and that of BTC/USDC $3.08 \to 6.37$, compared with $0.79 \to 1.02$ for BTC/USD.

Basis decomposition

$$D_t = m\left(\ln P_t^{\mathrm{BTC}/Q} - \ln P_t^{\mathrm{BTC/USD}}\right), \qquad \delta_t = m \ln P_t^{Q/\mathrm{USD}}, \qquad b_t = -D_t - \delta_t$$

- For stablecoin Q, we decompose the cross-currency basis D_t into the peg deviation δ_t and the triangular arbitrage deviation b_t.
- Triangular arbitrage parity, $P^{\mathrm{BTC/USD}} = P^{\mathrm{BTC}/Q} P^{Q/\mathrm{USD}}$, implies $b_t = 0$ (Reynolds et al., 2021), so full pass-through of the peg means $D_t = -\delta_t$.
- b_t is the part of the basis that the peg does not explain.
- The variance of the basis decomposes into

$$\mathrm{Var}(D) = \mathrm{Var}(\delta) + \mathrm{Var}(b) + 2\,\mathrm{Cov}(\delta, b).$$

Half-life of deviations

$$x_t = c + \rho x_{t-1} + u_t, \qquad \text{HL} = \frac{\ln 0.5}{\ln \rho}$$

- For $x \in \{\delta, b\}$, we estimate a first-order autoregressive model and the half-life HL in minutes (Lyons and Viswanath-Natraj, 2023).

Pass-through regression

$$\Delta b_t = \alpha + \beta b_{t-1} + \left(\lambda_0 + \boldsymbol{\lambda}_1^\top \tilde{\mathbf{z}}_{t-1}\right) \delta_{t-1} + \boldsymbol{\theta}^\top \tilde{\mathbf{z}}_{t-1} + \kappa \widetilde{\mathrm{RV}}_{t-1} + \varepsilon_t$$

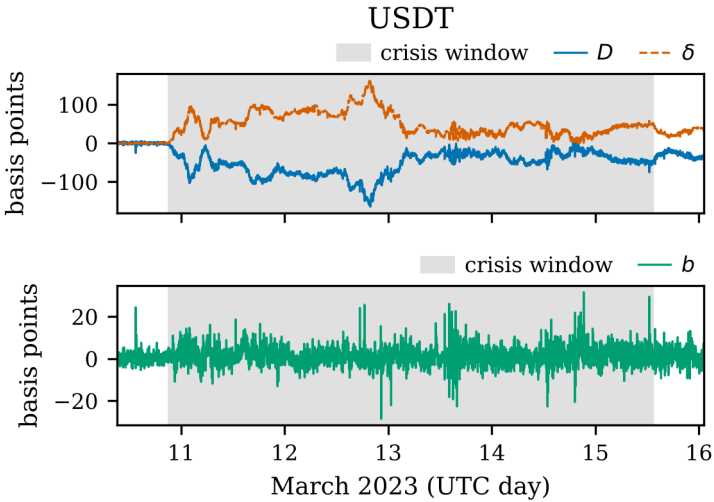
- CS is the spread estimator of Corwin and Schultz (2012), RV the realized volatility (Appendix A.1), and a tilde marks standardization within the estimation sample.
- The vector $\mathbf{z}_{t-1}$ holds the lagged CS and Kyle illiquidity of BTC/Q and the lagged Kyle illiquidity of Q/USD, and RV_{t-1} is the lagged RV of BTC/USD.
- $\boldsymbol{\theta}$ and κ are control coefficients, β measures the reversion speed of b, and λ_0 measures how the lagged peg deviation moves b at average liquidity.
- The vector $\boldsymbol{\lambda}_1$ measures how that response changes with a 1 standard deviation rise in each illiquidity measure, so liquidity governs the pass-through when $\boldsymbol{\lambda}_1 \neq \mathbf{0}$.

Spillover index and Granger causality

$$\mathbf{r}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{\Phi}_l \mathbf{r}_{t-l} + \varepsilon_t, \qquad \mathrm{E}\left[\varepsilon_t \varepsilon_t^\top\right] = \mathbf{\Sigma}, \qquad S = \frac{1}{5} \sum_{i \neq j} \tilde{\phi}_{ij}$$

- We model the joint dynamics of the five pairs by a vector autoregression (VAR) on 5-minute scaled log returns, with the lag order p is chosen by the BIC up to 6 lags.
- $\phi_{ij}(H)$ is the generalized share of the H-step forecast error variance of pair i due to shocks in pair j (Koop et al., 1996, Pesaran and Shin, 1998).
- The normalized shares $\tilde{\phi}_{ij}$ sum to 100 in each row (Appendix C.1), and S is the total spillover index of Diebold and Yilmaz (2012) with $H = 10$ (50 minutes).
- Pair j Granger-causes pair i (Granger, 1969) when the lags of j help predict i, which we test with a Wald test that all p coefficients of j in the equation of i are zero.

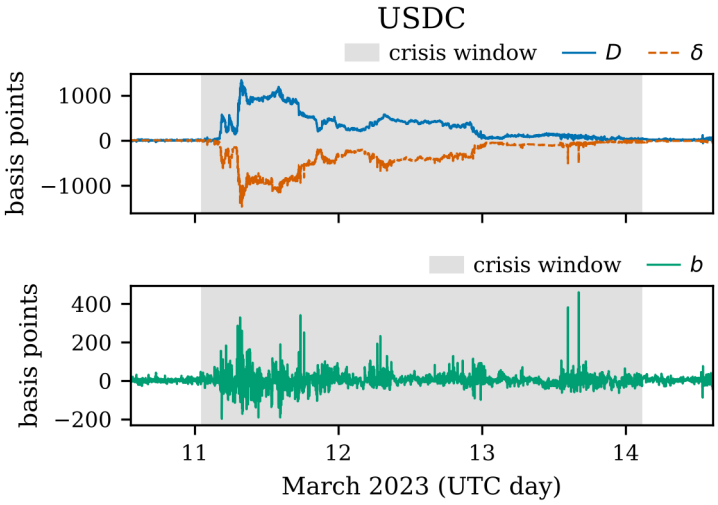
Decomposition of the basis, USDT



- Before the depeg, all three series stayed close to zero.
- Once the peg broke, D mirrored δ almost exactly, while b fluctuated around zero with short spikes.

Cross-currency basis D and peg deviation δ (top), and triangular arbitrage deviation b (bottom) in bps, 1-minute close prices from 12 hours before to 12 hours after each crisis window (shaded), UTC.

Decomposition of the basis, USDC



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Cross-currency basis D and peg deviation δ (top), and triangular arbitrage deviation b (bottom) in bps, 1-minute close prices from 12 hours before to 12 hours after each crisis window (shaded), UTC.

Decomposition on synchronous minutes

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
Mean D	-0.11	-53.2	-26.0	0.13	399	7.23
Mean δ	0.12	51.6	25.3	-0.61	-400	-7.17
$\mathrm{Var}(\delta)/\mathrm{Var}(D)$	0.56	1.02	0.97	0.67	1.00	0.63
Mean b	-0.01	1.57	0.70	0.47	1.72	-0.06
$\lvert b\rvert$ median	1.03	2.50	2.39	2.23	13.4	5.10
$\lvert b\rvert$ 99th percentile	5.37	12.3	10.9	17.3	140	26.5

Decomposition on synchronous minutes ($F_t = 1$), in bps. The remaining moments are in Appendix B.1.

- During the crisis, $\mathrm{corr}(D, -\delta) = 0.99$ for both stablecoins and $\mathrm{Var}(\delta)/\mathrm{Var}(D)$ is 1.02 for USDT and 1.00 for USDC, so δ accounts for almost the entire variance of D.
- The crisis widened the distribution of b, whose mean was below 2 bps, while for USDC the median of $|b|$ rose $2.23 \to 13.4$ bps.

Half-lives

		pre	crisis	post
USDT	δ	5.72	158	27.8
	b	0.45	0.82	0.64
USDC	δ	0.92	63.9	2.47
	b	0.43	0.52	0.61

Half-life in minutes on consecutive synchronous minutes ($FF_t = 1$).

- During the crisis, the half-life of b is 0.82 minutes for USDT and 0.52 minutes for USDC, compared with 158 and 63.9 minutes for δ.
- Arbitrage therefore halved triangular deviations within a minute even at the height of the stress, while the peg deviation persisted for hours.
- The persistence of the basis was hence the persistence of the peg.

Liquidity and pass-through

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
β	-0.810^{***}	-0.589^{***}	-0.688^{***}	-0.898^{***}	-0.744^{***}	-0.691^{***}
	(-13.77)	(-18.92)	(-23.47)	(-13.42)	(-12.84)	(-7.88)
λ_0	-0.0782^{***}	-0.0098^{***}	-0.0387^{**}	-0.7004^{***}	0.0015	0.0780
	(-2.82)	(-2.88)	(-2.53)	(-7.67)	(0.24)	(1.26)
λ_1, CS of BTC/Q	0.0161	-0.0019	-0.0096^{**}	0.2642^{***}	-0.0006	-0.0079
	(0.61)	(-1.02)	(-2.09)	(2.63)	(-0.13)	(-0.24)
λ_1, Kyle of BTC/Q	0.0124	-0.0045^{*}	0.0205^{**}	0.0735	0.0027	-0.0220
	(0.34)	(-1.65)	(2.20)	(1.08)	(1.06)	(-1.25)
λ_1, Kyle of Q/USD	0.0301^{*}	-0.0009	-0.0060	0.1195^{***}	0.0137^{***}	-0.0317
	(1.89)	(-0.31)	(-1.01)	(7.52)	(2.91)	(-1.06)
Joint p, $\lambda_1 = 0$	0.208	0.263	0.021	<0.001	0.007	0.568

Pass-through regression on consecutive synchronous minutes, with t statistics in parentheses.
*, ** and *** denote significance at 10%, 5% and 1%.

- During the crisis, the liquidity terms are jointly significant for USDC but not for USDT.
- At the mean discount of -400 bps, a 1 standard deviation rise in the Kyle illiquidity of USDC/USD implies a move in b of about -5.5 bps per minute.

Variance shares

	pre	crisis	post
Panel A. Variance shares (%)			
Share of BTC/USDC from BTC/USD	32.4	21.4	31.1
Share of BTC/USDC from USDC/USD	0.0	5.9	0.2
Share of USDC/USD from BTC/USDC	0.2	10.3	0.3

Spillovers of 5-minute returns over the union window, Panel A. Shares are the normalized $\tilde{\phi}_{ij}$ with $H = 10$.

- The total spillover index barely moved, at $39.7 \rightarrow 37.5 \rightarrow 39.4$.
- The crisis did not change overall connectedness but did change its composition.
- The share of BTC/USDC forecast error variance due to BTC/USD fell from 32.4% before the crisis to 21.4% during it, and USDC/USD took part of its place.

Granger causality

	pre	crisis	post
Panel B. Granger causality, Wald statistic			
BTC/USD → BTC/USDC	17.79***	10.11***	28.77***
BTC/USDC → BTC/USD	3.51	1.89	7.54
USDC/USD → BTC/USDC	1.37	14.87***	5.27
BTC/USDC → USDC/USD	6.00	114.19***	7.06
BTC/USD → BTC/USDT	16.76**	6.99**	29.54***
BTC/USDT → BTC/USD	10.06	13.57***	3.05
BTC/USDT → USDT/USD	28.45***	15.22***	28.42***
USDT/USD → BTC/USDT	5.15	0.03	4.84

Spillovers of 5-minute returns over the union window, Panel B. *, ** and *** denote significance at 10%, 5% and 1%.

- BTC/USD led BTC/USDC in every regime and never the reverse.
- USDC/USD and BTC/USDC led each other only during the crisis.
- During the crisis, BTC/USDC partly became a market for USDC itself, so that its prices and the USDC peg predicted each other.

Robustness of key crisis estimates

	USDC pass-through			Half-life, USDT / USDC		
Check	λ_1	Joint p	USDT λ_0	b	δ	Share
R0 baseline	0.0137^{***} (2.91)	0.007	-0.0098^{***} (-2.88)	0.82 / 0.52	158 / 63.9	21.4
R1 5-minute bars	0.0143^{*} (1.83)	0.106	-0.0132^{***} (-3.16)	2.47 / 2.27	284 / 219	–
R2 CS of BTC/USDC	0.0005 (0.11)	0.910	-0.0084^{**} (-2.36)	–	–	–
R2 Kyle of BTC/USDC	0.0040 (1.33)	0.185	-0.0091^{**} (-2.56)	–	–	–
R2 Kyle of USDC/USD	0.0149^{***} (2.93)	0.003	-0.0093^{***} (-2.70)	–	–	–

Key crisis estimates, with t statistics in parentheses. Share is the percentage of BTC/USDC forecast error variance due to BTC/USD. *, ** and *** denote significance at 10%, 5% and 1%.

- R1 re-estimates the half-lives and the pass-through regression on 5-minute bars, and R2 enters each liquidity measure alone.
- On 5-minute bars the coefficient on the Kyle illiquidity of USDC/USD stays similar but is significant only at 10%, with a joint p of 0.106.
- The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.

Conclusion

- We decomposed the cross-currency basis of USDT and USDC on 1-minute data and examined whether the response of b to δ depended on lagged illiquidity.
- During the crisis, δ carried almost the entire variance of D, and b had a half-life below 1 minute, against about 1 hour for the USDC peg and over 2.5 hours for the USDT peg.
- Liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.
- The stress that reached the stablecoin price of Bitcoin was the peg itself. During the crisis, BTC/USDC partly decoupled from BTC/USD and traded with the USDC peg.

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Code and replication data are available at *GitHub*.

Appendix A.1. Liquidity and volatility measures

Write $\overline{P}_t$ and $\underline{P}_t$ for the high and the low of a bar. The spread estimator of Corwin and Schultz (2012) over bars $t-1$ and t is defined by

$$\psi_t = \sum_{i=0}^{1} \left(\ln \frac{\overline{P}_{t-i}}{\underline{P}_{t-i}} \right)^2, \qquad \omega_t = \left(\ln \frac{\max(\overline{P}_t, \overline{P}_{t-1})}{\min(\underline{P}_t, \underline{P}_{t-1})} \right)^2,$$

$$\zeta_t = \frac{\sqrt{2\psi_t} - \sqrt{\psi_t}}{3 - 2\sqrt{2}} - \sqrt{\frac{\omega_t}{3 - 2\sqrt{2}}}, \qquad \mathrm{CS}_t = m \max \left\{ 0, \frac{2(e^{\zeta_t} - 1)}{1 + e^{\zeta_t}} \right\},$$

and the realized volatility is

$$\mathrm{RV}_t = m \left(\sum_{i=0}^{119} r_{t-i}^2 \right)^{1/2}.$$

Appendix B.1. Remaining decomposition moments

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
SD D	2.17	30.4	8.46	4.12	327	10.4
SD δ	1.63	30.7	8.31	3.36	327	8.28
SD b	1.80	4.08	3.78	5.21	38.3	8.83
$\mathrm{Var}(b)/\mathrm{Var}(D)$	0.68	0.02	0.20	1.60	0.01	0.72
$2\mathrm{Cov}(\delta, b)/\mathrm{Var}(D)$	-0.24	-0.04	-0.17	-1.27	-0.01	-0.35
$\lvert b \rvert$ 95th percentile	3.32	8.47	7.72	9.76	81.0	17.3

Remaining decomposition moments on synchronous minutes ($F_t = 1$), in bps.

Appendix B.2. Control coefficients and fit

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
θ, CS of BTC/Q	0.001	−0.011	0.273**	0.041	−1.895	−0.695
	(0.05)	(−0.11)	(2.01)	(0.40)	(−1.01)	(−1.05)
θ, Kyle of BTC/Q	0.025	0.315*	−0.537**	−0.033	0.366	−0.502*
	(0.70)	(1.79)	(−2.07)	(−0.26)	(0.25)	(−1.69)
θ, Kyle of Q/USD	0.016	−0.106	−0.075	−0.220	5.517***	−0.081
	(0.58)	(−0.70)	(−0.44)	(−1.48)	(2.86)	(−0.14)
κ	0.158**	−0.056	0.075	0.298*	0.989	1.442***
	(2.54)	(−0.44)	(0.76)	(1.79)	(0.86)	(2.76)
R^2	0.408	0.290	0.342	0.442	0.373	0.378

Control coefficients and fit of the pass-through regression, with t statistics in parentheses. *, ** and *** denote significance at 10%, 5% and 1%.

Appendix C.1. Spillover index

Write the VAR in its moving-average form,

$$\mathbf{r}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \mathbf{A}_h \boldsymbol{\varepsilon}_{t-h}, \qquad \mathbf{A}_0 = \mathbf{I}, \qquad \mathbf{A}_h = \sum_{l=1}^{\min(h,p)} \boldsymbol{\Phi}_l \mathbf{A}_{h-l}.$$

The generalized forecast error variance decomposition uses only $\mathbf{A}_h$ (Pesaran and Shin, 1998),

$$\phi_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{A}_h^\top \mathbf{e}_i}, \qquad \tilde{\phi}_{ij} = \frac{100\,\phi_{ij}}{\sum_l \phi_{il}},$$

- σ_{jj} is the jth diagonal element of $\boldsymbol{\Sigma}$, and $\mathbf{e}_i$ selects pair i.
- Each row is normalized so that $\sum_j \tilde{\phi}_{ij} = 100$, since the generalized shares of a row need not sum to 1.
- The total spillover index S is the average share of forecast error variance that comes from the other four pairs (Diebold and Yilmaz, 2012).

Appendix C.2. Granger causality

For Granger causality from pair j to pair i, let $\gamma_{ij} = ([\mathbf{\Phi}_1]_{ij}, \ldots, [\mathbf{\Phi}_p]_{ij})^\top$ collect the p coefficients of j in the equation of i. The null hypothesis $H_0 \colon \gamma_{ij} = \mathbf{0}$ is tested with

$$W_{ij} = \hat{\gamma}_{ij}^\top \left[\widehat{\mathrm{Var}} \left(\hat{\gamma}_{ij} \right) \right]^{-1} \hat{\gamma}_{ij} \ \sim \ \chi^2_p,$$

where $\widehat{\mathrm{Var}}(\hat{\gamma}_{ij})$ is the corresponding block of the estimated covariance of the VAR coefficients (Granger, 1969).